

LAB NO : 04

LAB TITLE : Switch Statement, Break and Continue in C++

OBJECTIVE :

The objective of this lab is to understand the use of the switch statement in C++. Students also learn how the break statement controls program flow and how the continue statement is used inside loops. This lab also introduces simple menu-driven programs using switch statements.

THEORY :

The switch statement is used when a program needs to select one option from multiple choices based on the value of a variable. It is often used instead of multiple if-else statements when comparing a single variable with several constant values. Each option inside the switch statement is written as a case. The program checks the value of the variable and executes the matching case block. The break statement is used to stop execution inside a case. Without break, the program would continue executing the next cases. The default statement is optional and runs when none of the case values match the given input. The continue statement is used inside loops to skip the current iteration and move to the next iteration.

SOFTWARE :

1. .DEV-C++ IDE
2. C++ Compiler

PROCEDURE / PROCESS :

Day Name Using Switch :

1. Start the program
2. Declare an integer variable for day number
3. Ask the user to enter a number from 1 to 7
4. Use a switch statement to check the number

5. Display the corresponding day name
6. End the program

```
#include <iostream> // input aur output ke liye library
using namespace std; // standard namespace use karne ke liye

int main()
{
    int day; // integer variable jisme day number store hoga

    cout << "Enter day number (1-7): "; // user se day number input lene ka
message
    cin >> day; // user se value lena

    switch(day) // day ke mutabiq case select hoga
    {
        case 1:
            cout << "Monday";
            break; // agar 1 ho to Monday

        case 2:
            cout << "Tuesday";
            break; // agar 2 ho to Tuesday

        case 3:
            cout << "Wednesday";
            break; // agar 3 ho to Wednesday

        case 4:
            cout << "Thursday";
            break; // agar 4 ho to Thursday

        case 5:
            cout << "Friday";
            break; // agar 5 ho to Friday
```

```

        case 6:
            cout << "Saturday";
            break; // agar 6 ho to Saturday

        case 7:
            cout << "Sunday";
            break; // agar 7 ho to Sunday

        default:
            cout << "Invalid day number";
            // agar 1-7 ke ilawa koi number ho to error message
    }

    return 0; // program end
}

```

Grade Evaluation Using Switch :

1. Start the program
2. Declare a character variable for grade
3. Ask the user to enter a grade
4. Use a switch statement to evaluate the grade
5. Display the corresponding message
6. End the program

```

#include <iostream> // input aur output ke liye library
#include <cctype>    // topper function ke liye library

using namespace std; // standard namespace

int main()
{
    char grade; // character variable jisme grade store hoga

    cout << "Enter your grade (A, B, C, D, F): "; // user se grade lena
    cin >> grade; // input lena
}

```

```

    grade = toupper(grade); // lowercase ko uppercase me convert karna

    switch(grade) // grade ke mutabiq decision lena
    {
        case 'A':
            cout << "Excellent"; // A grade ka result
            break;

        case 'B':
            cout << "Very Good"; // B grade ka result
            break;

        case 'C':
            cout << "Good"; // C grade ka result
            break;

        case 'D':
            cout << "Pass"; // D grade ka result
            break;

        case 'F':
            cout << "Fail"; // F grade ka result
            break;

        default:
            cout << "Invalid Grade"; // agar koi aur input ho
    }

    return 0; // program end
}

```

Unit Conversion Menu Using Switch :

1. Start the program

2. Display a menu of conversion options
3. Take user choice
4. Ask for input value
5. Perform the selected conversion
6. Display the result
7. End the program.

```
#include <iostream> // input aur output ke liye library

using namespace std; // standard namespace

int main()
{
    int choice; // user ki choice store karne ke liye
    double value; // value store karne ke liye (decimal bhi ho sakti hai)

    // menu display karna
    cout << "1 → Kilometers to Meters\n";
    cout << "2 → Meters to Centimeters\n";
    cout << "3 → Kilograms to Grams\n";
    cout << "4 → Celsius to Fahrenheit\n";
    cout << "Enter your choice: ";

    cin >> choice; // choice input lena

    cout << "Enter value: ";
    cin >> value; // value input lena

    switch(choice) // choice ke mutabiq conversion hoga
    {
        case 1:
            cout << "Meters = " << value * 1000; // kilometers ko meters me
            convert karna
            break;

        case 2:
```

```

        cout << "Centimeters = " << value * 100; // meters ko centimeters
me convert karna
        break;

    case 3:
        cout << "Grams = " << value * 1000; // kilograms ko grams me
convert karna
        break;

    case 4:
        cout << "Fahrenheit = " << (value * 9/5) + 32; // celsius ko
fahrenheit me convert karna
        break;

    default:
        cout << "Invalid choice"; // agar galat option ho
    }

    return 0; // program end
}

```

CONCLUSION :

In this lab we learned how to use the switch statement for multiple decision making in C++. We also understood the importance of the break statement in controlling program flow. This lab helped us develop simple menu-based programs and improved our understanding of conditional structures.